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PAPER

Systemic lupus erythematosus with and without a family history: a meta-analysis

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Objective: The objective of this paper is to investigate the association of clinical manifestations and laboratory parameters between familial systemic lupus erythematosus (SLE) and sporadic SLE. **Methods:** All relevant literature was retrieved from the PubMed, EMBASE, Web of Science and China National Knowledge Infrastructure (CNKI) databases. The qualities of these studies were evaluated using a modified version of the Newcastle-Ottawa scale. The characteristics and clinical manifestations of involved individuals were extracted from each study. Pooled odds ratio (OR) was calculated using the random effects-method, and the heterogeneity between studies was quantified using the I^2 statistic. **Results:** Of 330 studies identified by the search strategy, six were included in this review. In total, 733 cases were familial SLE and 1405 were sporadic SLE. Analysis revealed that photosensitivity, nephritis and thrombocytopenia were negatively associated with familial SLE, with OR (95% CI) values of 0.73 (0.60–0.89), 0.72 (0.59–0.88) and 0.75 (0.57–0.98), respectively. **Conclusions:** Photosensitivity, thrombocytopenia and renal involvement could be more common in non-familial SLE, which should be further confirmed by well-designed studies with large populations. *Lupus* (2017) **0,** 1–6.

Key words: Systemic lupus erythematosus; family history; meta-analysis

Introduction

Systemic lupus erythematosus (SLE) is a complex autoimmune disease characterized by diverse clinical phenotypes and various autoantibodies against nuclear components. The disease is believed to result from an interaction of genetic, epigenetic, hormonal and environmental factors.[1] The contribution of genetic factors to SLE development was evidenced by the high heritability (> 66%) and sibling risk ratio ($8 < \lambda s < 29$), and the higher concordance rate in monozygotic twins (24–69%) than in dizygotic twins or siblings (2–5%) among SLE as well.[2] Familial lupus accounts for about 10% of SLE cases and has long been appreciated as an appropriate model to study the interaction of genetic and environmental factors.[3] More than 40 susceptibility loci have been identified within the past several decades.[4] However, further studies are required to determine how the causal variants at each locus lead to an SLE-specific phenotype.[5] On the other hand, as a clinician, one of the most frequent questions asked by patients is whether the "healthy" family members will develop SLE and how their clinical manifestations, disease duration and prognosis are different from SLE patients without a family history. Thus, to better understand the association of genetic factors with the manifestations and the pathogenesis of SLE, we performed a meta-analysis to evaluate the correlation of the clinical manifestations and laboratory disorders in patients with familial and sporadic-onset SLE.

Methods

Literature search

A literature search to identify all relevant articles was conducted according to the search strategies published by Furlan et al.[6] The literature databases of PubMed, EMBASE, Web of Science and China National Knowledge Infrastructure (CNKI) were searched without language restrictions. The last search update was performed on March 30, 2017.

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The search strategies were based on the following terms: "lupus," "SLE," or "systemic lupus erythematosus," combined with "familial," "sporadic," "familial history," "families," or "genetic epidemiology." Thesaurus terms and free texts were used. The references in the identified or related articles were then manually reviewed to search other relevant citations.

Studies meeting the following criteria were included in this review: (a) studies evaluating the differences between familial lupus and sporadic lupus with clinical and serological manifestations; (b) cohort studies designed with familial lupus; and (c) studies with genetic epidemiology sufficient data available to calculate the odds ratio (OR) with 95% confidence interval (95% CI). Studies were excluded in any of the following cases: (a) not a human study; (b) not an original study; (c) cohort designs leading to outcome of interest and (d) study not pertaining to the clinical or immunologic features of familial SLE. We also compared the authors' names, affiliations and geographic locations and the periods of studies to identify sequential or multiple publications. If several articles were published by the same authors using the same patients' data, only studies with the largest sample size or the most recent study were included.

Two reviewers separately included papers using the inclusion and exclusion criteria. Relevant data from each eligible article was extracted independently by two authors using a standardized data extraction form: the first author's name, publication date, location, total number of SLE patients, sex ratio, the mean age at inclusion, the mean age onset, clinical manifestations (expressed as either the number or percentage of familial and sporadic patients). Discrepancies were resolved by group discussion.

Quality evaluation

The quality assessment of included studies was based on the Newcastle-Ottawa assessment (NOS) scale.[7] The criteria for selection assessment were modified according to the Critical Appraisal Skills Program (CASP)[8] and Quality Assessment Tool for Systematic Reviews of Observational Studies (QATSO)[9] to make the subcategories more applicable to grouped cohort studies. Total score varies from 0 to nine points, with nine representing the highest quality.

Statistical analysis

Mantel-Haenszel ORs with 95% CIs were used to assess the strength of the association of familial history with various clinical and immunological features.[10] The ORs for each study were calculated and then pooled in a meta-analysis using Review Manager (Rev Man) (computer program) version 5.3 (Copenhagen: The Nordic Cochrane Centre, The Cochrane Collaboration, 2011).

Both the chi-square-based Q-statistic test and I^2 were used to assess the heterogeneity among studies. The I^2 value represented the proportion of variation due to heterogeneities in studies. The heterogeneity level was classified into four categories: unimportant (0–40%), moderate (30–60%), substantial (50–90%) and considerable (75–100%) heterogeneity.[11] If the p value of the heterogeneity Q-statistic ≤ 0.10 or $I^2 \geq 50\%$, the random-effects model was selected.[12] Otherwise, a fixed-effects model was adopted.[13] Publication bias was investigated by funnel plot if there were more than 10 studies involved.

Results

Eligible studies and characteristics

As shown in Figure 1, the literature search identified 330 articles that were then screened for eligibility: A total of 194 were repeated citations, and 94 were deemed irrelevant based on a review of the title/abstract. The remaining 40 articles underwent a full text review, and an additional 34 articles were excluded: Twenty-five had no comparative data on familial and sporadic SLE, eight focused on a specialized subset of patients (only female patients or patients with kidney diseases), and one utilized the same study population. The remaining six articles[14–19] were selected for this meta-analysis (Table 1).

Study quality

The qualities of all involved studies were evaluated with the NOS and are listed in Table 2. Of the six studies, one scored six points, and two scored seven points, and the other scored nine points. All studies defined SLE cases with the American College of Rheumatology (ACR) criteria (1982 or 1997 revision) and described the characteristics of patients in that context. Data from all of SLE patients and their affected relatives were collected by reviewing their medical records if available or by personal interview and blood collection if the data were missing.

Comparison of familial and sporadic SLE

Table 1 lists the demographics of patients included in the retrieved studies. The gender ratio

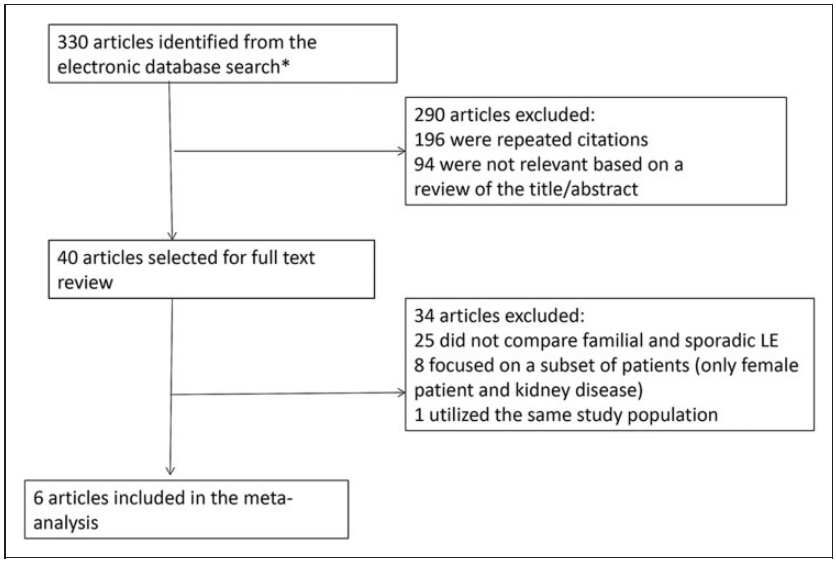

Figure 1 Flowchart of study selection process.
LE: lupus erythematosus.

Table 1 Characteristics of included studies

				Familial SLE vs sporadic SLE					
					Gender (F:M)				
Study ID	*Location*	*Total cases*	*Gender (F:M)*	*Cases*	*Familial*	*Sporadic*	*Mean age at inclusion*	*Mean onset ages*	*Disease duration*
Gourley et al. (1996)[14]	Ireland	68	60:8	34 vs 34	N/A	N/A	N/A	N/A	N/A
Koskenmies et al. (2001)[15]	Finland	226	212:14	113 vs 113	102:11	102:11	34.9 vs 34.3	30.7 vs 30.5	N/A
Michel et al. (2001)[16]	France	190	176:14	90 vs 100	83:7	93:7	38.7 vs 37.9	26.6 vs 26.2	N/A
Wang et al. (2007)[17]	China	695	630:65	50 vs 645	N/A	N/A	N/A	N/A	N/A
Sestak et al. (2008)[18,a]	America	896	813:83	425 vs 471	N/A	N/A	N/A	N/A	N/A
Abutiban et al. (2009)[19,a]	Kuwaiti	63	N/A	21 vs 42	N/A	N/A	28.5 vs 34	20.0 vs 26.5	8. 3 vs 7.1

N/A: not specified.
[a]The gender ratio was not different between familial and sporadic SLE (mentioned in text, with no numerical data).
SLE: systemic lupus erythematosus; F: female; M: male; ID: identification.

Table 2 Quality assessment with Newcastle-Ottawa assessment scale

	Study identification					
	Gourley et al. (1996)[14]	*Koskenmies et al. (2001)*[15]	*Michel et al.(2001)*[16]	*Wang et al. (2007)*[17]	*Sestak et al. (2008)*[18]	*Abutiban et al. (2009)*[19]
Selection						
Case cohort representative	*	*	*	*	*	*
Selection of non-exposed control	*	*	*	*	*	*
Ascertainment of exposure	*	*	*	*	*	*
Outcome negative at start	*	*	*	*	*	*
Comparability						
Disease course	–	–	–	–	–	*
Other factors	–	*	*	–	–	–
Outcome						
Outcome assessment	*	*	*	*	*	*
Duration of follow-up	–	*	*	–	–	*
Adequacy of follow-up	–	*	*	–	–	*
Total score	7	9	9	7	6	9

Maximum possible total score for case series is nine stars.

(female:male) varied from 7:1 to 15:1 among studies with no difference between familial and sporadic SLE as reported by four of six studies.[15,16,18,19] The mean ages at inclusion and the onset ages listed in three studies were comparable between patients with and without family history of SLE. In addition, Abutiban et al. analyzed disease duration and found no difference between the two groups.[19]

First, we analyzed the association of familial history with four subtypes of lupus mucocutaneous manifestations listed in the ACR classification criteria, including malar rash, discoid rash, oral ulcer and photosensitivity.[20] Six studies with 733 familial SLE cases and 1405 sporadic cases investigated the association of familial history with mucocutaneous disorders. The overall meta-analyses showed that familial SLE was negatively associated with photosensitivity (OR 0.73, 95% CI 0.60–0.89, $p = 0.002$) but with a moderate heterogeneity ($I^2 = 35\%$, $p_{\text{Heterogeneity}} = 0.17$) (Figure 2(a)). There were no significant differences in malar rash, discoid rash and oral ulcer between familial SLE cases and sporadic cases (Table 3).

Second, we analyzed the association of familial SLE with other clinical manifestations. Six studies, including 733 familial SLE cases and 1405 sporadic cases, evaluated the association of lupus nephritis with familial SLE. Meta-analysis revealed that familial SLE patients were less likely to have renal disorders than sporadic cases (OR 0.72, 95% CI 0.59–0.88, $p = 0.002$) (Figure 2(b)). Only four studies with 644 familial SLE cases and 722 sporadic cases investigated the association of familial history with thrombocytopenia. Meta-analysis revealed a negative association of familial SLE cases with thrombocytopenia (OR 0.75, 95% CI 0.57–0.98, $p = 0.04$) (Figure 2(c)). The heterogeneity among studies was not important in both analyses (lupus nephritis: $I^2 = 2\%$, $p_{\text{Heterogeneity}} = 0.4$; thrombocytopenia: $I^2 = 0\%$, $p_{\text{Heterogeneity}} = 0.6$). In addition, no associations of familial SLE with arthritis, serositis, neurological manifestation, alopecia, hemolytic anemia, Raynaud phenomenon, and anti-double-stranded DNA (anti-dsDNA) antibody were found by meta-analyses (Table 3).

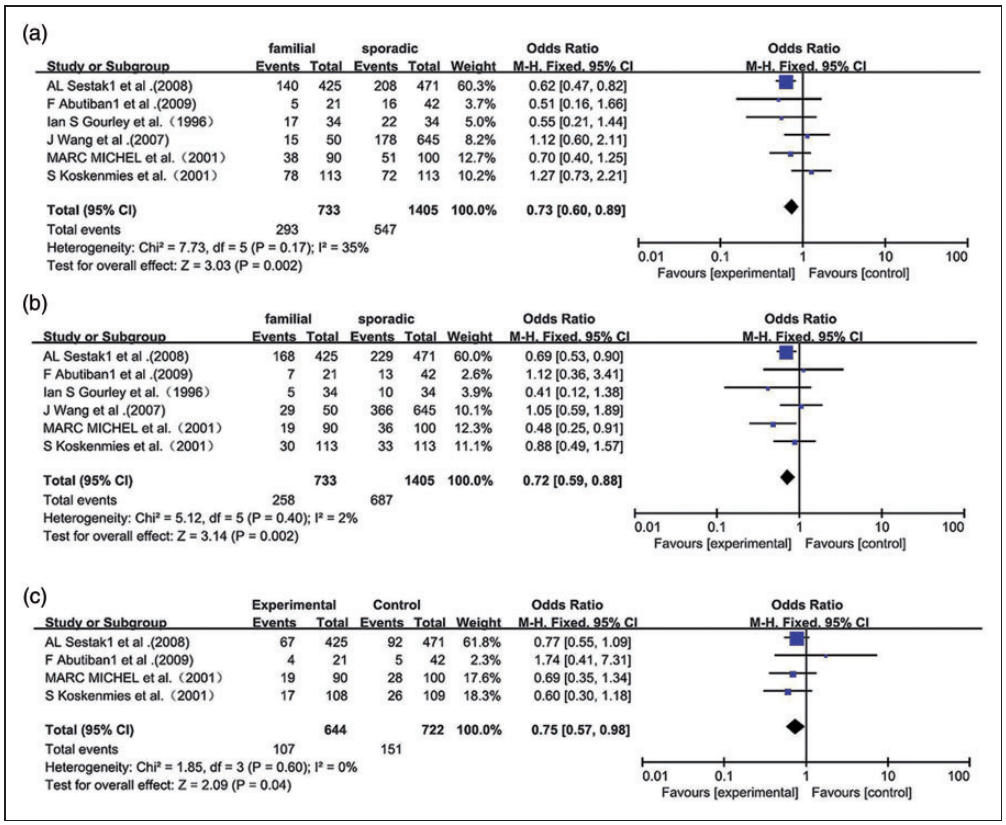

Figure 2 Forest plot of the association of familial SLE history with the clinical and laboratory parameters.
(a) The association of familial SLE history with photosensitivity. (b) The association of familial SLE history with renal involvement. (c) The association of familial SLE history with thrombocytopenia.
SLE: systemic lupus erythematosus; M-H: Mantel-Haenszel; OR: odds ratio; CI: confidence interval; df: degree of freedom.

Table 3 Meta-analysis of comparison between familial and sporadic systemic lupus erythematosus

	Study	*Case (familial:sporadic)*	*OR (95% CI)*	I^2	$p_{Heterogeneity}$	$p_{overall\ test}$
Malar rash	6	764:1405	1.06 (0.87, 1.28)	0%	0.85	0.58
Discoid rash	6	733:1405	0.97 (0.73, 1.29)	37%	0.16	0.85
Photosensitivity	6	733:1405	0.73 (0.60, 0.89)	35%	0.17	0.002
Arthritis	6	733:1405	1.23 (0.98, 1.54)	32%	0.19	0.07
Serositis	5	620:1292	0.86 (0.69, 1.08)	0%	0.73	0.20
Renal disorder	6	733:1405	0.72 (0.59, 0.88)	2%	0.40	0.002
Neurological manifestation	5	620:1292	1.12 (0.80, 1.56)	21%	0.28	0.51
Raynaud phenomenon	4	195:821	0.78 (0.52, 1.19)	0%	0.79	0.25
Alopecia	3	105:721	0.87 (0.54, 1.40)	0%	0.71	0.56
Hemolytic anemia	5	677:756	1.32 (0.90, 1.95)	0%	0.49	0.16
Thrombocytopenia	4	644:722	0.75 (0.57, 0.98)	0%	0.60	0.04
Anti-dsDNA	5	665:759	0.84 (0.67, 1.06)	12%	0.34	0.14
Oral ulcers	6	731:1405	1.36 (1.05, 1.76)	70%	0.005	0.02

Statistical model: Mantel-Haenszel (M-H), fixed; OR: odds ratio; CI: confidence interval; Anti-dsDNA: anti-double-stranded DNA.

Discussion

This study aimed to investigate the impact of familial history on the clinical manifestations and laboratory disorders in SLE patients. We found that photosensitivity, thrombocytopenia and renal involvement, but not the other clinical and laboratory parameters, were negatively associated with familial SLE.

The pathogenesis of SLE is multifactorial, including genetic and environmental factors as well as innate and adaptive immunological abnormalities. Some risk loci identified by genome-wide association studies (GWAS), such as ITGAM, FcγR, PRDM1-ATG5, and TNFAIP3, are strongly associated with SLE.[21] One genetic variant may be involved in several disease-related manifestations, suggesting a common pathogenic mechanism shared by a group of disorders. For instance, the genetic variants of ITGAM were found to be simultaneously associated with the involvement of skin, kidney and joints, and immunologic disorders as well.[22] Accordingly, we found that photosensitivity, thrombocytopenia and renal involvement were negatively associated with familial SLE, which could be related to some specific genetic variants in a common locus. In the course of SLE disease, more than 75% of patients have skin damage as one of the earliest symptoms and signs of SLE.[1,21,23] As for lupus, malar rash and discoid rash are two directly visible signs. The rashes lead to an early diagnosis and thus an early treatment of LE (including SLE, discoid LE and subacute cutaneous LE). Early treatment may reduce the risk of the development of renal and hematologic diseases, and decrease the risk of the transformation from cutaneous LE to SLE as well. In addition, patients have better compliance when patients and their families pay more attention to the disease. Thus, SLE may be well controlled at early onset without developing into severe symptoms in these patients. All together, these factors could contribute to the negative association of family SLE with photosensitivity, thrombocytopenia and renal involvement. In accordance with this finding, Wang et al. reported that class V lupus nephritis was less common in familial SLE, whereas Systemic Lupus Erythematosus Disease Index score and other clinic pathological features were not significantly different between 34 familial SLE and 102 sporadic SLE patients.[24]

There are several limitations to be considered when interpreting the results of this meta-analysis. First, with respect to the genetic component, the concordance rate was higher in monozygotic twins (>20%) than in dizygotic twins and other full siblings (2–5%), indicating a linkage between the degree of relatives and genetic factors.[25–27] In this meta-analysis, there were not enough data to analysis whether the lupus-related parameters varied with the degree of relatives. Second, the sex ratios were different in the included articles. Sexual hormones are critical for SLE development and associated with SLE manifestations. It was reported that female patients were highly associated with alopecia, photosensitivity, oral ulcers, arthritis, malar rash, lupus anticoagulant level, and low level of C3, while male patients were more likely to have renal involvement, serositis and

pleurisies, thrombocytopenia, and serum level of anti-dsDNA antibody.[28] Thus, the varied sexual ratios in each study could cause some bias. Third, the mean onset ages were not specified in three articles. The age of onset has been reported to be related to clinical manifestations. Patients with childhood-onset lupus were commonly found to have fever, malar rash, hematologic disorders (hemolytic anemia and thrombocytopenia), renal involvement and lymphadenopathy.[29] Last, it is important to note that a couple of the largest studies had lower quality but they would count as higher weight because of the large sizes.

Despite the limitations, this meta-analysis has many strong points. This is the only quantitative analysis of multiple studies comparing familial and non-familial SLE. Photosensitivity, thrombocytopenia and renal involvement could be more common in non-familial SLE compared with familial SLE.

Declaration of Conflicting Interests

The authors declared no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

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